

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Practicum 4: Reading and Managing Imaging Data
10/2/2026

Preamble for this Practicum and HWs 6 to 10:

In this and the next several Practicum and HW assignments, you will correct and make measurements from the data in `Files/hw/hw6/hw6_data/` on WebCourses. You can use the solutions to each assignment to correct your prior assignment, so that your errors don't propagate.

As always, if you have hard-coded values (e.g., filenames, constants for array sizes, etc.) they should be assigned to variables at the start of your main homework file. Do not hard-code any values into any functions you write. Pass the values as arguments to your functions, instead. Except at the highest level, you should write generic functions that do a task well, and pass in parameters to tune them for *this* task.

The data you will work with are from the WIRC camera at the Palomar 5-meter telescope. The original dataset was not quite FITS-compliant and needed a special program (`rdpharo.py`) to read them and turn them into normal, compliant FITS files. We have already done this work for you, but, to be intellectually honest, we also provide `rdpharo.py` and the original dataset. If you want to check it out, you can find it in the `hw6/archival_data` folder.

During this and the coming HW assignments, you will read FITS headers and data into Python arrays. Where possible, use the `+=`, `*=`, etc. operators to work on these arrays in place. (Why?)

It might be tempting to store FITS files in between steps. **Only** do this when requested, as otherwise the homework handins can get very large. Do it in your main homework file, not down in a function. The purpose of writing out these FITS files is to have something to grade, even if your code does not run in a fresh Python session. Do NOT read any FITS file that you have written, nor write one that was not requested.

Finally, if you are stuck, or if you see a problem and don't know how to fix it, and it is too late to get help from us, please describe the problem in comments in your homework file or log. Identifying a problem, even if you can't fix it, will give you partial credit!

1. As this is part of your HW, use your terminal to make a new folder `hw6_<yourname>` under your `ast4762/` homework folder. For this homework, your main homework file is a Python file named `hw6_<username>.py` (or `hw6_<username>.ipynb`, but remember to save it as a `.py` file before submission). Save it in your homework folder. You might want to keep some information in your log too.

2. **(10 points total)** Start with the minimal amount of hard-coded things at the start of your code.

a) Define a variable (e.g., `datadir`) that holds the data directory name (including the final `/`) and another variable (e.g., `fext`) that holds the FITS extension string (including the dot (as in `.fits`)). If you need more hard-coded values, come back and enter them in this section of the assignment.

b) Initialize two lists, one for your target images and one for your dark images (called, e.g., `objfile` and `darkfile`). These lists will contain the file names of the objects (files `stars_13s_*`) and dark (files `dark_13s_*`) files. To populate the lists, use the appropriate `os` command to list the data directory. Don't hard code the names in your homework file!

Your code should then identify whether a file should go into the `objfile` or `darkfile` list and populate the lists. Make sure to not include the `.fits` extension in the lists. We already have this under `fext`, or whatever variable you created, and we will be inserting some text between the root filename and the extension when we save processed files.

c) Use informative statements to print `datadir`, `fext`, and *only* the last elements of `objfile` and `darkfile` (one from each).

d) Read any one object file and determine the data array sizes. Assign these to two variables (e.g., `ny` and `nx`).

e) Define two variables (called, e.g., `nobj` and `ndark`) that contain the numbers of files in the `objfile` and `darkfile` lists by querying the sizes of the lists. Why should you not hardcode them?

Use informative statements to print `ny`, `nx`, `nobj`, and `ndark`.

3. **(10 points total)** Write a series of commands that makes and populates your data cubes with the target and dark images.

a) Create two 3D arrays (`float64`), one for the objects and one for the darks. Your arrays should have sizes `(nobj, ny, nx)` and `(ndark, ny, nx)` respectively. Print the shapes of these arrays.

b) Populate your data cubes. Read the target and dark data into the arrays. Save the last header from each set in a variable (e.g., `objhead` and `darkhead`).

Read and print the date of observation (`DATE-OBS`) from each of these.

c) **Extra credit** (5 points) Why not print `TIME-OBS`?